tropical: every campaign

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1 Status

Table 1 lists every campaign: the values its search tuned, that search's verdict (`tune/report/`), whether `frozen.yaml` is approved, and how much of its grid is stored. 13 of 13 campaigns have grid results.

Table 1: Per campaign: tuned values, tuning-report verdict (blocks/reviews), approval, grid runs stored/expected and diverged.

campaign	tuned	verdict	blk	rev	approved	runs	div.
Knapsack		–	0	0	**no**	240/240	0
SubsetSumDecision		–	0	0	**no**	240/240	0
MaxSubsetSum		–	0	0	**no**	240/240	0
FractionalKnapsack		–	0	0	**no**	240/240	0
MinCoinChange		–	0	0	**no**	240/240	0
Quickselect		–	0	0	**no**	240/240	0
BalancedPartition		–	0	0	**no**	220/240	0
BinPacking		–	0	0	**no**	216/240	0
ConvexHull		–	0	0	**no**	240/240	0
ThreeSumDecision		–	0	0	**no**	240/240	0
LIS		–	0	0	**no**	240/240	0
FloydWarshall		–	0	0	**no**	240/240	0
SCC		–	0	0	**no**	240/240	0

2 Results

Columns are the levels of `fishr_lambda`; each campaign's rows are the combinations of `model_type`, `head`. Means are over seeds, the best per row in bold. Δ is a level's paired difference against the control (0) by seed, marked * where $p < 0.05$ after the campaign's correction. Figure 1 shows the headline metric at a glance.

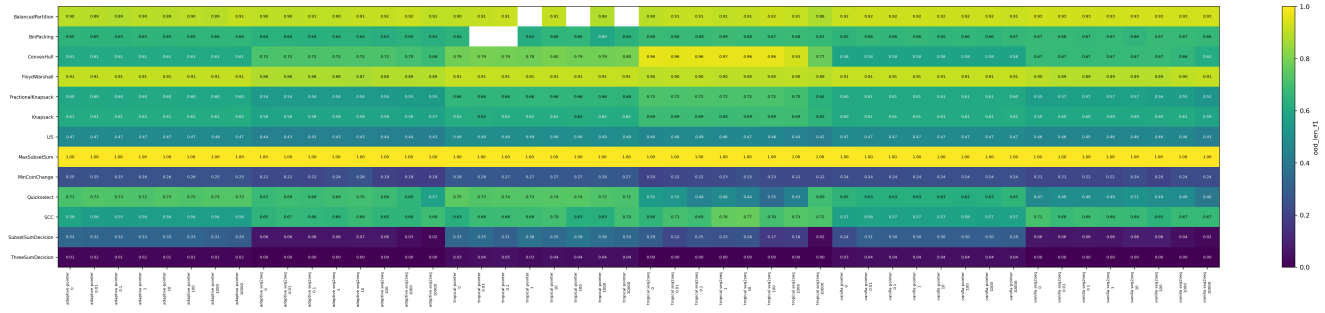


Figure 1: The headline metric, `ood_len_f1`: every campaign by cell.

2.1 `ood_len_f1` (higher is better)

Table 2: `ood_len_f1` per campaign: means by `fishr_l` against 0.

campaign	`model_type`	`head`	0	0.01	0.1	1	10	100	1000	10000
BalancedPartition	adaptive	pointer	0.897	0.892	0.885	0.896	0.892	0.9	0.893	**0.908**
BalancedPartition	adaptive	seq2seq	0.904	0.899	0.906	0.905	0.91	0.909	**0.917**	0.905
BalancedPartition	tropical	pointer	0.903	0.91	**0.915**	–	0.909	–	0.883	–
BalancedPartition	tropical	seq2seq	0.898	0.907	0.913	0.906	0.907	**0.919**	0.912	0.878
BalancedPartition	vanilla	pointer	0.917	0.922	0.921	0.922	0.922	0.921	**0.923**	0.914
BalancedPartition	vanilla	seq2seq	0.926	0.927	0.928	0.928	0.93	0.931	0.928	**0.932**
BinPacking	adaptive	pointer	0.65	0.647	0.647	0.654	0.651	0.655	**0.659**	0.658
BinPacking	adaptive	seq2seq	0.641	0.638	0.636	0.639	0.643	0.626	**0.646**	0.632
BinPacking	tropical	pointer	0.647	–	–	0.622	0.658	**0.662**	0.599	0.643
BinPacking	tropical	seq2seq	0.681	0.678	0.679	**0.685**	0.681	0.673	0.683	0.67
BinPacking	vanilla	pointer	0.649	0.665	0.657	0.664	0.654	0.661	0.656	**0.665**
BinPacking	vanilla	seq2seq	0.672	**0.677**	0.667	0.669	0.659	0.674	0.673	0.664
ConvexHull	adaptive	pointer	**0.611**	0.611	0.61	0.61	0.61	0.61	0.61	0.606
ConvexHull	adaptive	seq2seq	0.715	0.72	**0.72**	0.719	0.72	0.719	0.7	0.682
ConvexHull	tropical	pointer	0.785	0.795	0.792	0.782	**0.797**	0.787	0.787	0.796
ConvexHull	tropical	seq2seq	0.965	0.96	0.964	**0.965**	0.965	0.961	0.93	0.77
ConvexHull	vanilla	pointer	**0.579**	0.579	0.579	0.579	0.579	0.579	0.579	0.578
ConvexHull	vanilla	seq2seq	0.673	0.669	0.669	0.669	**0.674**	0.672	0.663	0.619
FloydWarshall	adaptive	pointer	0.907	0.909	0.909	0.909	0.909	**0.909**	0.909	0.908
FloydWarshall	adaptive	seq2seq	0.881	0.875	0.876	0.875	0.875	0.885	0.887	**0.889**
FloydWarshall	tropical	pointer	0.913	0.914	0.912	0.914	0.913	**0.914**	0.914	0.913
FloydWarshall	tropical	seq2seq	0.896	**0.904**	0.903	0.901	0.901	0.901	0.903	0.895
FloydWarshall	vanilla	pointer	0.911	0.914	0.914	0.914	**0.914**	0.913	0.913	0.913
FloydWarshall	vanilla	seq2seq	0.9	0.889	0.887	0.887	0.888	0.887	0.903	**0.912**
FractionalKnapsack	adaptive	pointer	0.601	0.602	0.601	0.602	**0.602**	0.601	0.597	0.598
FractionalKnapsack	adaptive	seq2seq	0.537	0.54	0.54	0.54	0.539	0.537	0.545	**0.545**
FractionalKnapsack	tropical	pointer	0.663	0.658	0.659	**0.664**	0.662	0.659	0.657	0.656
FractionalKnapsack	tropical	seq2seq	**0.721**	0.717	0.72	0.717	0.718	0.718	0.699	0.64
FractionalKnapsack	vanilla	pointer	0.605	**0.608**	0.608	0.608	0.607	0.607	0.606	0.604
FractionalKnapsack	vanilla	seq2seq	0.555	0.566	0.567	0.567	**0.572**	0.561	0.552	0.518
Knapsack	adaptive	pointer	0.613	0.614	0.614	0.615	0.613	0.615	0.614	**0.618**
Knapsack	adaptive	seq2seq	0.582	0.584	0.584	0.584	**0.585**	0.583	0.58	0.575
Knapsack	tropical	pointer	0.618	**0.623**	0.617	0.616	0.615	0.621	0.619	0.616
Knapsack	tropical	seq2seq	0.687	0.687	0.687	0.683	0.685	0.689	**0.69**	0.649
Knapsack	vanilla	pointer	0.604	0.611	0.611	0.611	0.61	0.611	0.613	**0.619**
Knapsack	vanilla	seq2seq	0.588	0.598	0.597	0.596	0.597	0.602	**0.61**	0.586
LIS	adaptive	pointer	0.473	0.475	0.474	0.475	0.474	0.475	**0.475**	0.474
LIS	adaptive	seq2seq	**0.436**	0.435	0.434	0.434	0.434	0.436	0.436	0.43
LIS	tropical	pointer	0.494	0.493	0.49	**0.495**	0.485	0.493	0.488	0.495
LIS	tropical	seq2seq	0.459	0.475	**0.478**	0.465	0.467	0.459	0.444	0.425
LIS	vanilla	pointer	0.474	0.472	0.473	0.474	0.473	0.473	0.474	**0.475**
LIS	vanilla	seq2seq	0.457	0.457	0.454	0.457	0.46	**0.462**	0.458	0.427
MaxSubsetSum	adaptive	pointer	**1**	1	1	1	1	1	1	1
MaxSubsetSum	adaptive	seq2seq	**1**	1	1	1	1	1	1	1
MaxSubsetSum	tropical	pointer	**1**	1	1	1	1	1	1	1
MaxSubsetSum	tropical	seq2seq	**1**	1	1	1	1	1	1	1
MaxSubsetSum	vanilla	pointer	**1**	1	1	1	1	1	1	1
MaxSubsetSum	vanilla	seq2seq	**1**	1	1	1	1	1	1	1
MinCoinChange	adaptive	pointer	0.25	0.255	0.255	0.256	**0.256**	0.256	0.255	0.255
MinCoinChange	adaptive	seq2seq	0.221	0.221	0.215	0.235	**0.26**	0.194	0.184	0.184
MinCoinChange	tropical	pointer	0.276	0.275	0.271	0.274	0.274	0.273	**0.282**	0.274
MinCoinChange	tropical	seq2seq	0.202	0.219	0.223	0.229	**0.246**	0.227	0.221	0.221
MinCoinChange	vanilla	pointer	0.24	0.242	0.242	0.243	0.243	0.242	0.243	**0.244**
MinCoinChange	vanilla	seq2seq	0.214	0.214	0.216	0.218	0.237	0.203	0.236	**0.237**
Quickselect	adaptive	pointer	0.714	0.729	0.726	0.722	0.731	0.723	0.733	**0.734**
Quickselect	adaptive	seq2seq	0.675	0.688	0.68	0.692	**0.7**	0.686	0.654	0.573
Quickselect	tropical	pointer	**0.749**	0.73	0.736	0.731	0.736	0.741	0.724	0.711

Table 2: `ood_len_f1` per campaign: means by `fishr_l` against 0.

campaign	`model_type`	`head`	0	0.01	0.1	1	10	100	1000	10000
Quickselect	tropical	seq2seq	0.547	0.52	0.443	0.48	0.44	0.349	0.432	**0.692**
Quickselect	vanilla	pointer	0.652	0.625	0.63	0.626	0.625	0.623	0.632	**0.655**
Quickselect	vanilla	seq2seq	0.469	0.49	0.492	0.488	**0.513**	0.491	0.488	0.396
SCC	adaptive	pointer	**0.59**	0.558	0.551	0.559	0.559	0.557	0.557	0.559
SCC	adaptive	seq2seq	0.652	0.667	0.662	0.656	0.684	0.653	0.655	**0.69**
SCC	tropical	pointer	0.631	0.681	0.684	0.689	**0.702**	0.628	0.627	0.7
SCC	tropical	seq2seq	0.656	0.707	0.688	0.761	**0.769**	0.702	0.733	0.72
SCC	vanilla	pointer	0.568	0.576	0.573	0.573	0.573	**0.58**	0.57	0.574
SCC	vanilla	seq2seq	**0.709**	0.679	0.69	0.683	0.678	0.647	0.674	0.67
SubsetSumDecision	adaptive	pointer	**0.328**	0.322	0.323	0.325	0.327	0.327	0.305	0.294
SubsetSumDecision	adaptive	seq2seq	0.06	0.0617	0.0603	0.0599	**0.0656**	0.0629	0.0316	0.0157
SubsetSumDecision	tropical	pointer	0.328	0.353	0.307	0.375	0.349	**0.387**	0.356	0.333
SubsetSumDecision	tropical	seq2seq	**0.295**	0.217	0.254	0.248	0.261	0.165	0.181	0.0242
SubsetSumDecision	vanilla	pointer	0.239	**0.307**	0.302	0.304	0.304	0.297	0.3	0.293
SubsetSumDecision	vanilla	seq2seq	0.0636	0.0609	0.0631	0.0611	**0.0792**	0.0786	0.0395	0.0199
ThreeSumDecision	adaptive	pointer	0.0145	0.0151	0.0122	**0.0161**	0.0146	0.0133	0.0149	0.0151
ThreeSumDecision	adaptive	seq2seq	0	0	0	0	0	0	0	3.74×10^{-5}
ThreeSumDecision	tropical	pointer	0.0337	0.0399	**0.0496**	0.0321	0.0442	0.0382	0.0406	0.0357
ThreeSumDecision	tropical	seq2seq	0	**0.0032**	3.30×10^{-4}	0.00169	0.00267	0	0	0
ThreeSumDecision	vanilla	pointer	0.0348	0.0394	**0.0411**	0.0405	0.0407	0.0393	0.0387	0.039
ThreeSumDecision	vanilla	seq2seq	0	1.10×10^{-4}	2.97×10^{-4}	3.71×10^{-5}	0	0	2.22×10^{-4}	0

A Other metrics

A.1 ood_len_exact (higher is better)

Table 3: ood_len_exact per campaign: means by fishr against 0.

campaign	model_type	head	0	0.01	0.1	1	10	100	1000	10000
BalancedPartition	adaptive	pointer	0.0208	0.0198	0.0199	0.0212	0.0207	0.0204	**0.0234**	0.0226
BalancedPartition	adaptive	seq2seq	0.0173	0.0176	0.0184	0.0171	0.019	0.019	**0.0206**	0.0136
BalancedPartition	tropical	pointer	**0.0257**	0.0234	0.0236	–	0.0243	–	0.0184	–
BalancedPartition	tropical	seq2seq	0.0143	0.0153	0.0154	0.0144	0.0161	**0.0176**	0.0146	0.0106
BalancedPartition	vanilla	pointer	0.0247	0.0239	0.0246	0.0249	0.0235	0.0236	**0.0255**	0.021
BalancedPartition	vanilla	seq2seq	0.0229	0.0222	0.023	0.0223	0.0228	**0.0233**	0.022	0.023
BinPacking	adaptive	pointer	0.0707	0.0693	0.073	0.0727	0.0709	**0.0742**	0.0732	0.0728
BinPacking	adaptive	seq2seq	0.0587	0.0588	0.0587	**0.0624**	0.0622	0.0575	0.0604	0.0442
BinPacking	tropical	pointer	0.0886	–	–	**0.0953**	0.0878	0.0883	0.0928	0.093
BinPacking	tropical	seq2seq	0.0824	0.0831	**0.087**	0.0836	0.082	0.0814	0.0824	0.0792
BinPacking	vanilla	pointer	0.0776	0.0793	**0.0797**	0.0758	0.0778	0.0774	0.078	0.0759
BinPacking	vanilla	seq2seq	0.0659	0.0606	0.0646	**0.0659**	0.0649	0.0651	0.0644	0.0597
ConvexHull	adaptive	pointer	**0.0881**	0.0865	0.0864	0.0859	0.0865	0.0859	0.0867	0.0833
ConvexHull	adaptive	seq2seq	0.191	0.195	**0.195**	0.193	0.194	0.194	0.158	0.115
ConvexHull	tropical	pointer	0.377	0.381	0.381	0.386	**0.404**	0.387	0.402	0.383
ConvexHull	tropical	seq2seq	0.654	0.642	0.665	**0.67**	0.663	0.649	0.499	0.216
ConvexHull	vanilla	pointer	0.0743	0.074	0.0737	0.0737	0.074	0.0736	0.0751	**0.0758**
ConvexHull	vanilla	seq2seq	0.143	0.138	0.138	0.138	**0.145**	0.143	0.121	0.0652
FloydWarshall	adaptive	pointer	0.053	0.0524	0.0516	**0.0532**	0.053	0.0521	0.0527	0.0509
FloydWarshall	adaptive	seq2seq	0.0509	0.0553	0.0551	0.0554	0.0553	0.0533	**0.0558**	0.0549
FloydWarshall	tropical	pointer	0.0566	0.0567	0.0565	0.0568	0.0568	0.0567	**0.0568**	0.0566
FloydWarshall	tropical	seq2seq	0.0261	**0.0332**	0.0301	0.0281	0.0294	0.0314	0.0292	0.0277
FloydWarshall	vanilla	pointer	0.0518	0.0553	0.0553	**0.0555**	0.0551	0.0546	0.0553	0.0546
FloydWarshall	vanilla	seq2seq	0.0504	0.0556	0.0556	0.0555	0.0553	0.0556	0.0559	**0.0566**
FractionalKnapsack	adaptive	pointer	0.0579	0.0611	0.0603	**0.0613**	0.0604	0.0599	0.0566	0.0558
FractionalKnapsack	adaptive	seq2seq	0.0414	0.044	0.0448	0.0447	0.0443	**0.047**	0.0339	0.0271
FractionalKnapsack	tropical	pointer	0.112	0.111	0.111	0.111	**0.113**	0.111	0.111	0.104
FractionalKnapsack	tropical	seq2seq	0.163	0.163	**0.163**	0.16	0.161	0.159	0.141	0.0907
FractionalKnapsack	vanilla	pointer	0.0497	0.0503	**0.0504**	0.0495	0.0503	0.0496	0.0493	0.0491
FractionalKnapsack	vanilla	seq2seq	0.0259	0.0317	0.0312	0.032	**0.0339**	0.0277	0.0243	0.0146
Knapsack	adaptive	pointer	0.0436	0.0453	0.0445	0.0449	0.0444	0.0444	0.0443	**0.0471**
Knapsack	adaptive	seq2seq	0.0288	0.029	0.0288	0.0287	0.0295	0.0279	0.0307	**0.0312**
Knapsack	tropical	pointer	0.0409	**0.0452**	0.041	0.0439	0.0414	0.0437	0.0429	0.0419
Knapsack	tropical	seq2seq	0.0734	**0.0765**	0.0748	0.073	0.0745	0.0743	0.0732	0.0553
Knapsack	vanilla	pointer	0.0333	0.0359	0.0359	0.036	0.0352	0.0363	0.0374	**0.0399**
Knapsack	vanilla	seq2seq	0.0272	0.0325	0.0329	0.0332	**0.0332**	0.0322	0.0332	0.0313
LIS	adaptive	pointer	0.0132	0.0138	0.0136	0.014	0.0137	0.0139	**0.0146**	0.0145
LIS	adaptive	seq2seq	0.0044	0.00427	0.00417	0.00427	0.00427	**0.00563**	0.00443	0.003
LIS	tropical	pointer	0.0618	0.0676	**0.0692**	0.064	0.065	0.0617	0.0646	0.0649
LIS	tropical	seq2seq	0.0533	**0.0541**	0.05	0.0516	0.0538	0.0521	0.05	0.0391
LIS	vanilla	pointer	0.0119	0.0119	**0.0126**	0.0125	0.0122	0.0124	0.0126	0.012
LIS	vanilla	seq2seq	0.00883	0.00823	0.00847	0.00873	**0.00897**	0.00853	0.00647	0.00307
MaxSubsetSum	adaptive	pointer	**1**	1	1	1	1	1	1	1
MaxSubsetSum	adaptive	seq2seq	**1**	1	1	1	1	1	1	1
MaxSubsetSum	tropical	pointer	**1**	1	1	1	1	1	1	1
MaxSubsetSum	tropical	seq2seq	**1**	1	1	1	1	1	1	1
MaxSubsetSum	vanilla	pointer	**1**	1	1	1	1	1	1	1
MaxSubsetSum	vanilla	seq2seq	**1**	1	1	1	1	1	1	1
MinCoinChange	adaptive	pointer	**0.0521**	0.0496	0.0492	0.0499	0.0495	0.0502	0.0498	0.0511
MinCoinChange	adaptive	seq2seq	**0.0446**	0.0417	0.0421	0.0381	0.0445	0.0283	0.0135	0.0135
MinCoinChange	tropical	pointer	0.0502	**0.0523**	0.0506	0.0513	0.0501	0.0485	0.0489	0.05
MinCoinChange	tropical	seq2seq	0.0583	0.0574	0.0615	**0.0624**	0.0484	0.0358	0.0184	0.0184

Table 3: `ood_len_exact` per campaign: means by `fishr`
against 0.

campaign	model_type	head	0	0.01	0.1	1	10	100	1000	10000
MinCoinChange	vanilla	pointer	0.0444	0.0448	0.0452	0.0454	0.0452	**0.0455**	0.045	0.0446
MinCoinChange	vanilla	seq2seq	**0.033**	0.0299	0.0319	0.0299	0.0317	0.0299	0.0145	0.0146
Quickselect	adaptive	pointer	0.414	**0.425**	0.423	0.418	0.418	0.416	0.416	0.419
Quickselect	adaptive	seq2seq	0.331	0.352	0.34	0.362	**0.369**	0.356	0.303	0.197
Quickselect	tropical	pointer	**0.263**	0.181	0.205	0.195	0.21	0.229	0.206	0.179
Quickselect	tropical	seq2seq	0.231	**0.256**	0.215	0.247	0.225	0.217	0.2	0.255
Quickselect	vanilla	pointer	**0.397**	0.394	0.393	0.393	0.393	0.395	0.391	0.397
Quickselect	vanilla	seq2seq	0.235	0.22	0.227	0.23	**0.245**	0.242	0.229	0.159
SCC	adaptive	pointer	0.00435	6.50×10^{-4}	1.50×10^{-4}	0.00655	3.00×10^{-4}	1.50×10^{-4}	0.0024	**0.0131**
SCC	adaptive	seq2seq	0.0859	4.00×10^{-4}	5.00×10^{-4}	0.0049	0	7.00×10^{-4}	0.004	**0.126**
SCC	tropical	pointer	0.0715	0.0977	0.155	0.117	**0.202**	0.0138	0.012	0.124
SCC	tropical	seq2seq	0.105	0.0835	0.024	0.128	**0.175**	0.0994	0.107	0.129
SCC	vanilla	pointer	**0.014**	5.50×10^{-4}	0.00315	0	9.00×10^{-4}	0	0	1.50×10^{-4}
SCC	vanilla	seq2seq	0.00455	0.0056	0.0171	0.0052	**0.0521**	0.0243	0.01	0.0319
SubsetSumDecision	adaptive	pointer	0.533	0.533	0.533	0.533	0.533	0.532	0.532	**0.534**
SubsetSumDecision	adaptive	seq2seq	0.535	0.537	0.537	0.537	**0.537**	0.535	0.523	0.514
SubsetSumDecision	tropical	pointer	0.528	0.526	0.528	0.528	0.528	**0.529**	0.528	0.528
SubsetSumDecision	tropical	seq2seq	0.533	**0.535**	0.533	0.533	0.533	0.532	0.525	0.519
SubsetSumDecision	vanilla	pointer	0.532	0.532	0.532	0.532	0.532	**0.533**	0.532	0.532
SubsetSumDecision	vanilla	seq2seq	0.535	0.536	0.535	0.536	**0.537**	0.535	0.526	0.515
ThreeSumDecision	adaptive	pointer	**0.402**	0.402	0.402	0.402	0.402	0.402	0.402	0.402
ThreeSumDecision	adaptive	seq2seq	0.403	**0.403**	0.403	0.403	0.403	0.403	0.403	0.403
ThreeSumDecision	tropical	pointer	**0.403**	0.402	0.402	0.402	0.403	0.403	0.403	0.402
ThreeSumDecision	tropical	seq2seq	**0.403**	0.403	0.403	0.403	0.403	0.403	0.403	0.403
ThreeSumDecision	vanilla	pointer	**0.402**	0.402	0.402	0.402	0.402	0.402	0.402	0.402
ThreeSumDecision	vanilla	seq2seq	**0.403**	0.403	0.403	0.403	0.403	0.403	0.403	0.403

A.2 `ood_val_f1` (higher is better)

Table 4: `ood_val_f1` per campaign: means by `fishr_lambda`
against 0.

campaign	model_type	head	0	0.01	0.1	1	10	100	1000	10000	Δ 0.01
BalancedPartition	adaptive	pointer	0.429	0.428	0.456	0.447	0.433	0.435	0.441	**0.46**	-6.1×10^{-4}
BalancedPartition	adaptive	seq2seq	0.437	0.408	0.391	0.435	0.451	**0.456**	0.377	0.28	-0.029
BalancedPartition	tropical	pointer	0.55	0.545	**0.576**	–	0.51	–	0.527	–	-0.0044
BalancedPartition	tropical	seq2seq	0.219	0.195	0.209	0.199	0.221	0.0735	0.174	**0.267**	-0.024
BalancedPartition	vanilla	pointer	0.539	0.541	0.521	0.56	0.522	0.557	0.516	**0.571**	0.0014
BalancedPartition	vanilla	seq2seq	0.353	0.451	0.411	0.49	**0.539**	0.434	0.426	0.436	0.098
BinPacking	adaptive	pointer	0.981	**0.993**	0.986	0.99	0.989	0.976	0.983	0.973	0.0015
BinPacking	adaptive	seq2seq	**0.999**	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0
BinPacking	tropical	pointer	0.996	–	–	0.998	0.997	0.997	**1**	0.998	–
BinPacking	tropical	seq2seq	0.999	0.998	0.998	0.999	0.999	**0.999**	0.998	0.998	-6.5×10^{-4}
BinPacking	vanilla	pointer	**0.993**	0.971	0.989	0.979	0.974	0.988	0.991	0.979	-0.014
BinPacking	vanilla	seq2seq	0.999	0.999	0.999	**0.999**	0.999	0.999	0.999	0.998	0
ConvexHull	adaptive	pointer	0.768	0.781	0.78	0.786	0.783	0.781	0.789	**0.804**	0.013
ConvexHull	adaptive	seq2seq	0.859	0.861	0.861	0.861	0.859	0.86	0.865	**0.866**	0.002
ConvexHull	tropical	pointer	0.806	0.809	0.778	0.795	0.788	**0.813**	0.785	0.792	0.0035
ConvexHull	tropical	seq2seq	0.803	0.815	0.807	0.812	**0.82**	0.795	0.785	0.8	0.012
ConvexHull	vanilla	pointer	**0.78**	0.761	0.762	0.761	0.764	0.766	0.741	0.708	-0.019
ConvexHull	vanilla	seq2seq	0.854	0.857	0.857	0.857	0.857	0.859	0.862	**0.867**	0.0029
FloydWarshall	adaptive	pointer	0.998	0.997	0.998	0.999	0.998	0.998	0.998	**0.999**	-5.5×10^{-4}
FloydWarshall	adaptive	seq2seq	0.197	0.35	**0.387**	0.31	0.347	0.212	0.2	0.232	0.15
FloydWarshall	tropical	pointer	0.97	**0.999**	0.996	0.87	0.987	0.804	0.957	0.92	0.016
FloydWarshall	tropical	seq2seq	0.536	0.846	0.633	0.775	0.771	0.413	**0.933**	0.916	0.22

Table 4: `ood_val_f1` per campaign: means by `fishr_lambda`, against 0.

campaign	`model_type`	`head`	0	0.01	0.1	1	10	100	1000	10000	Δ 0.01
FloydWarshall	vanilla	pointer	0.991	**0.996**	0.992	0.99	0.988	0.991	0.989	0.995	0.0048
FloydWarshall	vanilla	seq2seq	0.216	0.387	0.344	0.313	0.336	**0.402**	0.282	0.175	0.18
FractionalKnapsack	adaptive	pointer	**0.888**	0.864	0.873	0.874	0.875	0.874	0.869	0.86	-0.024
FractionalKnapsack	adaptive	seq2seq	0.896	0.89	0.892	0.89	0.896	0.895	**0.903**	0.89	-0.0059
FractionalKnapsack	tropical	pointer	0.857	0.858	0.866	**0.874**	0.87	0.865	0.864	0.857	0.0014
FractionalKnapsack	tropical	seq2seq	0.859	0.861	**0.871**	0.863	0.869	0.854	0.859	0.857	0.0024
FractionalKnapsack	vanilla	pointer	0.913	0.914	0.914	0.914	0.914	**0.917**	0.916	0.913	0.0014
FractionalKnapsack	vanilla	seq2seq	0.876	0.876	0.881	0.881	0.882	0.883	**0.892**	0.88	5.4×10^{-4}
Knapsack	adaptive	pointer	0.85	0.856	0.853	0.856	0.856	0.854	**0.857**	0.854	0.0062
Knapsack	adaptive	seq2seq	**0.856**	0.85	0.853	0.847	0.842	0.849	0.847	0.853	-0.005
Knapsack	tropical	pointer	**0.817**	0.811	0.814	0.813	0.807	0.805	0.817	0.803	-0.0058
Knapsack	tropical	seq2seq	0.798	0.787	0.794	0.797	**0.798**	0.793	0.791	0.788	-0.011
Knapsack	vanilla	pointer	0.851	0.855	0.854	0.853	0.854	0.852	0.852	**0.858**	0.0041
Knapsack	vanilla	seq2seq	0.85	0.855	**0.856**	0.856	0.855	0.847	0.85	0.854	0.0045
LIS	adaptive	pointer	0.728	**0.73**	0.722	0.73	0.728	0.724	0.724	0.724	0.0023
LIS	adaptive	seq2seq	0.682	0.672	0.67	0.67	**0.689**	0.68	0.673	0.65	-0.01
LIS	tropical	pointer	**0.674**	0.61	0.619	0.657	0.641	0.622	0.644	0.642	-0.064
LIS	tropical	seq2seq	0.618	0.653	**0.654**	0.592	0.603	0.623	0.573	0.528	0.035
LIS	vanilla	pointer	**0.654**	0.632	0.63	0.624	0.634	0.627	0.615	0.63	-0.022
LIS	vanilla	seq2seq	0.657	0.676	**0.682**	0.678	0.675	0.662	0.63	0.652	0.02
MaxSubsetSum	adaptive	pointer	**0.982**	0.981	0.981	0.981	0.981	0.981	0.981	0.981	0
MaxSubsetSum	adaptive	seq2seq	**1**	1	1	1	1	1	1	1	0
MaxSubsetSum	tropical	pointer	**0.982**	0.982	0.982	0.982	0.982	0.982	0.982	0.982	0
MaxSubsetSum	tropical	seq2seq	**1**	1	1	1	1	1	1	1	0
MaxSubsetSum	vanilla	pointer	**0.982**	0.981	0.981	0.981	0.981	0.981	0.981	0.98	0
MaxSubsetSum	vanilla	seq2seq	**1**	1	1	1	1	1	1	1	0
MinCoinChange	adaptive	pointer	**0.279**	0.259	0.273	0.27	0.271	0.265	0.271	0.267	-0.0075
MinCoinChange	adaptive	seq2seq	0.0922	**0.114**	0.111	0.0991	0.0734	0.0649	0	0	-0.0015
MinCoinChange	tropical	pointer	0.181	**0.313**	0.218	0.191	0.234	0.218	0.216	0.235	0.13
MinCoinChange	tropical	seq2seq	0.0479	0.106	0.0868	0.0593	**0.127**	0.00748	0.0107	0.0107	0.028
MinCoinChange	vanilla	pointer	0.24	0.244	0.246	0.238	**0.256**	0.252	0.256	0.246	0.007
MinCoinChange	vanilla	seq2seq	0.0367	0.0272	0.0277	0.0669	0.0404	**0.111**	0	0	3.9×10^{-4}
Quickselect	adaptive	pointer	0.996	0.996	0.996	0.996	0.996	0.996	0.996	**0.996**	1.3×10^{-4}
Quickselect	adaptive	seq2seq	**0.934**	0.933	0.933	0.933	0.933	0.932	0.928	0.918	2.4×10^{-5}
Quickselect	tropical	pointer	0.933	0.936	0.937	0.935	0.936	0.935	**0.938**	0.936	0.0025
Quickselect	tropical	seq2seq	0.752	0.749	0.753	0.751	**0.756**	0.743	0.724	0.68	-0.0031
Quickselect	vanilla	pointer	0.995	0.995	0.995	0.995	0.996	0.996	0.995	**0.996**	5.0×10^{-4}
Quickselect	vanilla	seq2seq	0.935	0.935	0.935	**0.935**	0.935	0.934	0.926	0.915	1.9×10^{-4}
SCC	adaptive	pointer	0.971	0.971	0.971	0.971	**0.971**	0.971	0.971	0.971	2.9×10^{-4}
SCC	adaptive	seq2seq	0.971	0.974	0.974	0.974	0.974	0.974	**0.974**	0.974	0.0026^{*}
SCC	tropical	pointer	0.971	0.971	0.969	0.971	**0.973**	0.971	0.973	0.971	-4.3×10^{-5}
SCC	tropical	seq2seq	0.979	0.978	0.978	0.978	0.976	0.977	**0.98**	0.979	-2.6×10^{-4}
SCC	vanilla	pointer	0.97	0.971	0.971	**0.971**	0.971	0.971	0.971	0.971	0.0011
SCC	vanilla	seq2seq	0.973	0.975	0.975	0.975	**0.975**	0.975	0.975	0.974	0.0022^{*}
SubsetSumDecision	adaptive	pointer	0.27	0.269	0.268	0.269	**0.274**	0.269	0.272	0.26	-7.6×10^{-4}
SubsetSumDecision	adaptive	seq2seq	0.144	0.151	0.147	0.138	**0.184**	0.154	0.0533	0.135	0.0071
SubsetSumDecision	tropical	pointer	0.301	0.315	0.328	0.327	**0.336**	0.309	0.323	0.302	0.014
SubsetSumDecision	tropical	seq2seq	0.282	0.328	0.297	0.309	**0.341**	0.293	0.315	0.108	0.046
SubsetSumDecision	vanilla	pointer	0.251	0.266	**0.266**	0.265	0.266	0.266	0.266	0.261	0.015
SubsetSumDecision	vanilla	seq2seq	**0.195**	0.179	0.19	0.17	0.174	0.132	0.0596	0.158	-0.016
ThreeSumDecision	adaptive	pointer	0.0175	0.0162	0.0133	0.0203	**0.021**	0.0178	0.0177	0.0205	1.4×10^{-5}
ThreeSumDecision	adaptive	seq2seq	7.48×10^{-4}	0	0	0	0	0	0	0	0
ThreeSumDecision	tropical	pointer	0.00457	0.00385	0.00595	**0.011**	0.00582	0.00809	0.00822	0.00387	0
ThreeSumDecision	tropical	seq2seq	0	0.00332	0	0	0	**0.011**	0.0052	0.00364	0
ThreeSumDecision	vanilla	pointer	**0.013**	0.00837	0.0102	0.00996	0.00997	0.00996	0.00977	0.00931	-0.0011
ThreeSumDecision	vanilla	seq2seq	0	0	0	0	0	**0.00106**	0	0	0

A.3 ood_val_exact (higher is better)

Table 5: ood_val_exact per campaign: means by fishr_lambda, a
against 0.

campaign	model_type	head	0	0.01	0.1	1	10	100	1000	10000	Δ 0.01	
BalancedPartition	adaptive	pointer	0.0105	0.00456	0.00828	0.00432	0.00664	0.00508	0.00792	**0.0146**	-6.0×10^{-4}	2.0
BalancedPartition	adaptive	seq2seq	0.0115	0.00972	0.00626	0.00886	**0.0156**	0.0124	0.00778	0.00122	-0.0019	
BalancedPartition	tropical	pointer	0.0117	0.0155	**0.0174**	–	0.0128	–	0.0143	–	-0.0042	
BalancedPartition	tropical	seq2seq	0.00572	0.00608	0.00396	0.0066	**0.0115**	0.00104	0.0048	0.00808	0	
BalancedPartition	vanilla	pointer	0.012	0.0139	0.0146	0.0166	0.0115	0.0126	**0.0231**	0.0145	5.0×10^{-4}	
BalancedPartition	vanilla	seq2seq	0.00974	0.0154	0.0139	0.0175	0.0149	0.0191	0.0097	**0.0205**	0.0011	−6.0
BinPacking	adaptive	pointer	0.866	0.928	0.923	0.933	**0.938**	0.822	0.883	0.874	0.063	
BinPacking	adaptive	seq2seq	**0.988**	0.987	0.983	0.987	0.987	0.987	0.988	0.987	-0.0015	
BinPacking	tropical	pointer	0.98	–	–	0.975	0.964	0.97	**0.995**	0.98	–	
BinPacking	tropical	seq2seq	0.99	0.979	0.979	0.988	0.985	**0.991**	0.978	0.981	-0.011	
BinPacking	vanilla	pointer	**0.945**	0.704	0.915	0.851	0.808	0.899	0.932	0.853	-0.24*	
BinPacking	vanilla	seq2seq	0.987	0.987	0.987	**0.987**	0.987	0.987	0.987	0.981	-3.6×10^{-4}	−4.0
ConvexHull	adaptive	pointer	0.101	0.111	0.11	0.117	0.113	0.109	0.121	**0.145**	0.011	
ConvexHull	adaptive	seq2seq	0.218	0.221	0.221	0.221	0.218	0.219	0.228	**0.23**	0.0033	
ConvexHull	tropical	pointer	0.14	0.15	0.0969	0.119	0.108	**0.152**	0.1	0.121	0.0097	
ConvexHull	tropical	seq2seq	0.137	0.155	0.142	0.148	**0.16**	0.118	0.103	0.132	0.018	
ConvexHull	vanilla	pointer	0.12	0.116	0.117	0.117	0.118	**0.122**	0.0943	0.0602	-0.0043	
ConvexHull	vanilla	seq2seq	0.21	0.214	0.213	0.215	0.213	0.217	0.225	**0.233**	0.0043	
FloydWarshall	adaptive	pointer	0.994	0.989	0.988	**0.996**	0.992	0.993	0.991	0.995	-0.0011	
FloydWarshall	adaptive	seq2seq	0.0493	0.142	**0.164**	0.129	0.122	0.0479	0.0292	0.0221	0.033	
FloydWarshall	tropical	pointer	0.86	**0.993**	0.972	0.679	0.891	0.604	0.821	0.814	0.1	
FloydWarshall	tropical	seq2seq	0.482	0.555	0.491	0.638	0.524	0.247	0.715	**0.821**	0	
FloydWarshall	vanilla	pointer	0.956	**0.982**	0.957	0.946	0.956	0.953	0.945	0.972	0.035	
FloydWarshall	vanilla	seq2seq	0.089	0.235	0.225	0.202	0.2	**0.254**	0.0945	0.0183	0.18	
FractionalKnapsack	adaptive	pointer	**0.36**	0.313	0.327	0.332	0.329	0.339	0.326	0.318	-0.048	
FractionalKnapsack	adaptive	seq2seq	0.365	0.351	0.355	0.35	0.367	0.363	**0.382**	0.342	-0.014	
FractionalKnapsack	tropical	pointer	0.257	0.287	0.286	**0.299**	0.29	0.286	0.285	0.27	0.03	
FractionalKnapsack	tropical	seq2seq	0.267	0.27	**0.292**	0.275	0.287	0.254	0.272	0.266	0.0034	
FractionalKnapsack	vanilla	pointer	0.425	0.429	0.428	0.428	0.428	**0.438**	0.434	0.426	0.0045	
FractionalKnapsack	vanilla	seq2seq	0.305	0.306	0.317	0.316	0.319	0.321	**0.336**	0.312	0.0014	
Knapsack	adaptive	pointer	0.321	0.319	0.315	0.32	0.319	0.315	**0.325**	0.313	-0.0017	
Knapsack	adaptive	seq2seq	**0.324**	0.317	0.318	0.314	0.295	0.311	0.318	0.299	-0.0069	
Knapsack	tropical	pointer	**0.219**	0.199	0.206	0.205	0.188	0.188	0.208	0.192	-0.019	
Knapsack	tropical	seq2seq	0.139	0.127	0.143	0.147	**0.15**	0.133	0.142	0.12	-0.013	
Knapsack	vanilla	pointer	0.317	0.32	0.315	0.314	0.317	0.311	0.313	**0.325**	0.0031	
Knapsack	vanilla	seq2seq	0.303	0.32	**0.324**	0.322	0.317	0.306	0.313	0.3	0.017	
LIS	adaptive	pointer	**0.262**	0.254	0.246	0.257	0.251	0.247	0.245	0.241	-0.0078	
LIS	adaptive	seq2seq	0.127	0.123	0.117	0.111	0.134	**0.142**	0.139	0.114	-0.0041	
LIS	tropical	pointer	**0.195**	0.158	0.165	0.185	0.192	0.171	0.179	0.179	-0.036	
LIS	tropical	seq2seq	0.0991	0.0971	0.0921	0.0923	0.092	0.0874	**0.104**	0.0792	-0.002	
LIS	vanilla	pointer	0.186	0.191	**0.196**	0.189	0.192	0.189	0.188	0.185	0.0057	
LIS	vanilla	seq2seq	0.154	**0.185**	0.184	0.174	0.18	0.166	0.12	0.132	0.031	
MaxSubsetSum	adaptive	pointer	**0.979**	0.972	0.972	0.972	0.973	0.972	0.97	0.97	0	
MaxSubsetSum	adaptive	seq2seq	**1**	1	1	1	1	1	1	1	0	
MaxSubsetSum	tropical	pointer	**0.98**	0.98	0.98	0.98	0.98	0.98	0.98	0.98	0	
MaxSubsetSum	tropical	seq2seq	**1**	1	1	1	1	1	1	1	0	
MaxSubsetSum	vanilla	pointer	**0.975**	0.972	0.972	0.971	0.972	0.971	0.97	0.968	0	
MaxSubsetSum	vanilla	seq2seq	**1**	1	1	1	1	1	1	1	0	
MinCoinChange	adaptive	pointer	0.163	0.122	0.122	**0.165**	0.158	0.121	0.124	0.138	-0.041	
MinCoinChange	adaptive	seq2seq	0.0773	0.0882	0.0837	0.0889	0.148	0.145	**0.493**	0.493	0.011	
MinCoinChange	tropical	pointer	0.312	**0.373**	0.345	0.302	0.339	0.329	0.311	0.35	0.062	
MinCoinChange	tropical	seq2seq	0.0595	0.111	0.067	0.0791	0.0856	0.143	0.353	**0.353**	0.052	
MinCoinChange	vanilla	pointer	0.145	0.19	0.194	0.187	**0.216**	0.206	0.215	0.186	0.045	
MinCoinChange	vanilla	seq2seq	0.0556	0.0533	0.0534	0.0673	0.0694	0.075	**0.501**	0.501	-0.0023	
Quickselect	adaptive	pointer	0.989	0.989	0.989	0.989	0.989	0.989	0.989	**0.99**	1.4×10^{-4}	3.2

Table 5: `ood_val_exact` per campaign: means by `fishr_lambda`, a
against 0.

campaign	model_type	head	0	0.01	0.1	1	10	100	1000	10000	Δ 0.01	
Quickselect	adaptive	seq2seq	**0.855**	0.854	0.854	0.854	0.854	0.853	0.847	0.819	-0.0011	-
Quickselect	tropical	pointer	0.841	0.847	0.849	0.842	0.846	0.845	**0.85**	0.846	0.0054	
Quickselect	tropical	seq2seq	**0.617**	0.607	0.602	0.617	0.612	0.595	0.568	0.506	-0.011	
Quickselect	vanilla	pointer	0.987	0.988	0.988	0.988	0.988	**0.988**	0.988	0.988	0.0013	
Quickselect	vanilla	seq2seq	0.856	0.857	0.857	**0.857**	0.857	0.855	0.843	0.813	0.0014	9.0
SCC	adaptive	pointer	0.885	0.885	0.886	0.885	0.886	0.885	0.884	**0.886**	4.4×10^{-4}	8.6
SCC	adaptive	seq2seq	0.881	0.899	0.899	0.902	0.9	0.903	**0.903**	0.896	0.018	
SCC	tropical	pointer	0.903	0.9	0.887	0.899	**0.909**	0.888	0.898	0.903	-0.0027	
SCC	tropical	seq2seq	**0.949**	0.941	0.936	0.944	0.935	0.924	0.944	0.948	-0.0084	
SCC	vanilla	pointer	0.878	0.888	0.885	0.888	0.885	0.886	0.887	**0.889**	0.0091	
SCC	vanilla	seq2seq	0.894	0.91	0.91	0.91	**0.911**	0.91	0.91	0.903	0.016	
SubsetSumDecision	adaptive	pointer	0.347	0.344	0.349	0.348	0.342	0.348	0.341	**0.363**	-0.0026	
SubsetSumDecision	adaptive	seq2seq	0.497	0.494	0.498	0.499	0.481	0.487	**0.514**	0.429	-0.0035	8.4
SubsetSumDecision	tropical	pointer	0.25	0.244	0.241	0.238	0.229	**0.261**	0.233	0.259	-0.0057	-
SubsetSumDecision	tropical	seq2seq	0.291	0.281	0.305	0.307	0.246	0.288	0.257	**0.455**	-0.0099	
SubsetSumDecision	vanilla	pointer	0.323	0.328	0.323	0.331	0.328	0.326	**0.331**	0.328	0.0054	4.4
SubsetSumDecision	vanilla	seq2seq	0.463	0.473	0.467	0.474	0.449	0.485	**0.509**	0.415	0.01	
ThreeSumDecision	adaptive	pointer	0.901	0.903	**0.925**	0.889	0.902	0.907	0.901	0.897	0.0026	
ThreeSumDecision	adaptive	seq2seq	0.969	**0.976**	0.976	0.976	0.976	0.976	0.976	0.976	0.0073	
ThreeSumDecision	tropical	pointer	0.963	0.966	0.96	0.946	0.963	**0.968**	0.948	0.966	0.0031	-
ThreeSumDecision	tropical	seq2seq	0.976	0.964	**0.976**	0.976	0.976	0.945	0.947	0.965	-0.013	
ThreeSumDecision	vanilla	pointer	0.876	**0.888**	0.887	0.881	0.882	0.883	0.884	0.884	0.011	
ThreeSumDecision	vanilla	seq2seq	0.975	0.976	0.976	**0.976**	0.976	0.975	0.973	0.976	9.6×10^{-4}	9.2

A.4 `ood_noise_f1` (higher is better)

Table 6: `ood_noise_f1` per campaign: means by `f`
against 0.

campaign	model_type	head	0	0.01	0.1	1	10	100	1000
BalancedPartition	adaptive	pointer	0.566	0.57	0.572	0.571	0.551	0.566	0.559
BalancedPartition	adaptive	seq2seq	0.649	0.619	0.602	0.642	0.601	0.647	**0.652**
BalancedPartition	tropical	pointer	0.588	0.593	0.572	–	0.557	–	**0.615**
BalancedPartition	tropical	seq2seq	**0.482**	0.398	0.468	0.37	0.369	0.302	0.354
BalancedPartition	vanilla	pointer	0.52	0.543	0.559	0.538	**0.567**	0.535	0.564
BalancedPartition	vanilla	seq2seq	0.612	0.61	0.621	0.639	**0.655**	0.6	0.577
BinPacking	adaptive	pointer	0.565	0.567	0.587	0.584	**0.588**	0.567	0.555
BinPacking	adaptive	seq2seq	0.62	0.615	**0.622**	0.621	0.619	0.621	0.619
BinPacking	tropical	pointer	0.628	–	–	0.631	0.633	0.623	**0.634**
BinPacking	tropical	seq2seq	0.611	0.624	0.626	0.622	**0.631**	0.623	0.625
BinPacking	vanilla	pointer	0.609	0.577	0.596	0.588	0.566	0.597	0.577
BinPacking	vanilla	seq2seq	**0.617**	0.61	0.607	0.613	0.603	0.6	0.604
ConvexHull	adaptive	pointer	0.84	0.839	0.839	0.839	0.839	0.839	0.84
ConvexHull	adaptive	seq2seq	0.86	0.86	0.86	0.86	0.86	0.86	0.861
ConvexHull	tropical	pointer	0.853	0.853	0.853	0.853	0.852	**0.854**	0.852
ConvexHull	tropical	seq2seq	0.857	0.858	0.858	**0.859**	0.858	0.857	0.855
ConvexHull	vanilla	pointer	**0.834**	0.832	0.833	0.832	0.832	0.832	0.831
ConvexHull	vanilla	seq2seq	0.859	0.86	0.86	0.86	0.86	0.86	0.861
FloydWarshall	adaptive	pointer	0.614	0.611	0.611	0.609	0.611	0.615	0.612
FloydWarshall	adaptive	seq2seq	0.623	0.624	0.624	**0.625**	0.624	0.62	0.613
FloydWarshall	tropical	pointer	0.588	0.61	0.592	0.597	**0.613**	0.585	0.605
FloydWarshall	tropical	seq2seq	**0.635**	0.625	0.618	0.612	0.614	0.618	0.623
FloydWarshall	vanilla	pointer	**0.618**	0.615	0.616	0.614	0.615	0.614	0.61
FloydWarshall	vanilla	seq2seq	0.594	0.601	0.601	**0.603**	0.602	0.601	0.591
FractionalKnapsack	adaptive	pointer	0.86	0.86	0.86	0.86	0.859	0.861	0.86

Table 6: ood_noise_f1 per campaign: means by f
against 0.

campaign	model_type	head	0	0.01	0.1	1	10	100	1000
FractionalKnapsack	adaptive	seq2seq	0.846	0.85	0.849	0.849	0.848	**0.852**	0.843
FractionalKnapsack	tropical	pointer	0.869	0.869	0.869	0.869	0.869	**0.871**	0.868
FractionalKnapsack	tropical	seq2seq	0.857	0.856	0.854	**0.861**	0.859	0.859	0.861
FractionalKnapsack	vanilla	pointer	0.859	0.86	0.86	0.86	0.86	0.86	**0.861**
FractionalKnapsack	vanilla	seq2seq	**0.856**	0.855	0.855	0.854	0.854	0.85	0.845
Knapsack	adaptive	pointer	0.652	0.656	0.651	0.649	0.654	0.655	0.652
Knapsack	adaptive	seq2seq	0.583	0.592	0.587	0.57	0.597	**0.627**	0.577
Knapsack	tropical	pointer	0.681	0.67	0.685	**0.687**	0.676	0.659	0.676
Knapsack	tropical	seq2seq	0.488	0.583	0.56	0.536	0.554	**0.594**	0.579
Knapsack	vanilla	pointer	0.62	0.628	0.622	**0.638**	0.626	0.624	0.619
Knapsack	vanilla	seq2seq	0.556	0.576	0.583	0.583	0.579	**0.589**	0.573
LIS	adaptive	pointer	0.622	**0.634**	0.631	0.633	0.629	0.621	0.623
LIS	adaptive	seq2seq	0.627	0.656	0.65	0.656	**0.677**	0.617	0.595
LIS	tropical	pointer	0.589	0.593	0.577	**0.599**	0.584	0.568	0.589
LIS	tropical	seq2seq	**0.587**	0.545	0.551	0.396	0.417	0.478	0.378
LIS	vanilla	pointer	**0.557**	0.531	0.521	0.517	0.526	0.524	0.515
LIS	vanilla	seq2seq	0.536	0.537	0.559	0.557	0.566	0.519	0.534
MaxSubsetSum	adaptive	pointer	**0.892**	0.887	0.887	0.887	0.887	0.887	0.887
MaxSubsetSum	adaptive	seq2seq	**1**	1	1	1	1	1	1
MaxSubsetSum	tropical	pointer	0.889	0.889	0.889	**0.894**	0.889	0.889	0.889
MaxSubsetSum	tropical	seq2seq	0.993	**0.999**	0.999	0.999	0.999	0.999	0.999
MaxSubsetSum	vanilla	pointer	**0.888**	0.887	0.887	0.887	0.887	0.887	0.887
MaxSubsetSum	vanilla	seq2seq	**1**	1	1	1	1	1	1
MinCoinChange	adaptive	pointer	0.302	0.299	0.302	**0.303**	0.3	0.302	0.3
MinCoinChange	adaptive	seq2seq	0.234	0.227	0.236	**0.239**	0.197	0.17	0.0916
MinCoinChange	tropical	pointer	0.321	0.326	**0.326**	0.325	0.322	0.323	0.321
MinCoinChange	tropical	seq2seq	0.267	0.299	**0.305**	0.276	0.276	0.208	0.133
MinCoinChange	vanilla	pointer	**0.284**	0.275	0.277	0.276	0.277	0.276	0.277
MinCoinChange	vanilla	seq2seq	**0.219**	0.206	0.202	0.208	0.21	0.172	0.11
Quickselect	adaptive	pointer	0.411	0.421	0.42	0.42	0.421	0.421	0.425
Quickselect	adaptive	seq2seq	0.43	**0.447**	0.447	0.446	0.447	0.446	0.444
Quickselect	tropical	pointer	0.544	0.541	0.526	0.527	**0.546**	0.543	0.537
Quickselect	tropical	seq2seq	0.461	0.485	0.481	0.489	0.472	0.474	**0.49**
Quickselect	vanilla	pointer	0.406	0.423	0.422	**0.423**	0.422	0.423	0.421
Quickselect	vanilla	seq2seq	0.432	0.441	0.441	0.441	0.442	0.448	**0.448**
SCC	adaptive	pointer	0.514	0.473	0.472	0.458	0.483	0.491	**0.546**
SCC	adaptive	seq2seq	0.559	**0.566**	0.544	0.551	0.564	0.555	0.545
SCC	tropical	pointer	0.591	0.609	**0.772**	0.635	0.681	0.614	0.636
SCC	tropical	seq2seq	0.652	0.656	0.659	0.684	**0.736**	0.714	0.634
SCC	vanilla	pointer	0.455	0.497	**0.501**	0.489	0.422	0.487	0.465
SCC	vanilla	seq2seq	0.508	0.599	0.584	0.56	0.548	0.568	**0.63**
SubsetSumDecision	adaptive	pointer	0.224	0.24	**0.242**	0.239	0.237	0.24	0.229
SubsetSumDecision	adaptive	seq2seq	0.141	0.114	0.109	0.128	**0.161**	0.13	0.0503
SubsetSumDecision	tropical	pointer	0.282	**0.294**	0.289	0.291	0.286	0.282	0.289
SubsetSumDecision	tropical	seq2seq	0.209	0.186	0.168	0.212	0.23	0.241	**0.244**
SubsetSumDecision	vanilla	pointer	**0.252**	0.234	0.232	0.237	0.238	0.235	0.224
SubsetSumDecision	vanilla	seq2seq	0.136	0.121	0.137	0.116	**0.148**	0.136	0.09
ThreeSumDecision	adaptive	pointer	0.00568	0.00463	0.0069	0.00511	0.00545	**0.00703**	0.00453
ThreeSumDecision	adaptive	seq2seq	0	0	0	2.25×10^{-4}	0	0	0
ThreeSumDecision	tropical	pointer	7.76×10^{-4}	**0.00182**	3.26×10^{-4}	1.11×10^{-4}	5.17×10^{-4}	0	5.44×10^{-4}
ThreeSumDecision	tropical	seq2seq	0	6.48×10^{-4}	3.22×10^{-4}	**0.00707**	0.00349	0.00192	5.50×10^{-4}
ThreeSumDecision	vanilla	pointer	**0.019**	0.0139	0.0138	0.0142	0.0145	0.0142	0.0138
ThreeSumDecision	vanilla	seq2seq	0	0	0	0	0	0	0

A.5 ood_noise_exact (higher is better)

Table 7: ood_noise_exact per campaign: means by fishr_lamb
Δ against 0.

campaign	model_type	head	0	0.01	0.1	1	10	100	1000	10000	Δ 0.01
BalancedPartition	adaptive	pointer	**0.0421**	0.0413	0.0383	0.0409	0.0383	0.0394	0.0399	0.0402	-8.4×10^{-4}
BalancedPartition	adaptive	seq2seq	**0.0314**	0.0284	0.027	0.0199	0.0269	0.0288	0.0225	0.0201	-0.003
BalancedPartition	tropical	pointer	0.0378	**0.041**	0.0378	–	0.0365	–	0.0374	–	0.0032
BalancedPartition	tropical	seq2seq	**0.0271**	0.0208	0.0248	0.0206	0.0149	0.0173	0.0182	0.0226	-0.0062
BalancedPartition	vanilla	pointer	0.032	0.0325	0.0342	0.0342	0.0363	0.0323	**0.0374**	0.0317	4.4×10^{-4}
BalancedPartition	vanilla	seq2seq	**0.0344**	0.0279	0.0326	0.0274	0.0266	0.0285	0.0232	0.0302	-0.0065
BinPacking	adaptive	pointer	0.0715	0.0617	0.0722	0.0694	0.0662	0.0649	0.0631	**0.0727**	-0.0098
BinPacking	adaptive	seq2seq	0.06	0.0581	0.0615	0.0606	0.0589	**0.0641**	0.0594	0.0614	-0.0019
BinPacking	tropical	pointer	0.0674	–	–	0.0674	0.0697	0.0637	**0.0706**	0.0663	–
BinPacking	tropical	seq2seq	0.0565	0.0599	0.0595	**0.0658**	0.0621	0.0598	0.0618	0.0636	0.0034
BinPacking	vanilla	pointer	0.0773	0.0599	0.066	0.0587	0.0698	0.0617	0.0642	**0.0783**	-0.017
BinPacking	vanilla	seq2seq	**0.0566**	0.0481	0.0479	0.051	0.0485	0.0428	0.0467	0.0543	-0.0084
ConvexHull	adaptive	pointer	0.217	0.217	0.217	0.217	0.217	0.216	0.219	**0.224**	3.2×10^{-4}
ConvexHull	adaptive	seq2seq	0.255	0.255	0.256	0.256	0.254	0.255	0.257	**0.258**	8.2×10^{-4}
ConvexHull	tropical	pointer	0.269	**0.271**	0.268	0.268	0.267	0.27	0.267	0.267	0.0026
ConvexHull	tropical	seq2seq	0.274	0.278	0.278	**0.279**	0.278	0.276	0.271	0.247	0.0038
ConvexHull	vanilla	pointer	**0.206**	0.205	0.205	0.205	0.205	0.205	0.205	0.197	-0.0015
ConvexHull	vanilla	seq2seq	0.249	0.251	0.251	0.251	0.25	0.25	**0.252**	0.248	0.0015
FloydWarshall	adaptive	pointer	**0.00658**	0.00638	0.00628	0.00658	0.0065	0.00638	0.00616	0.00642	-1.0×10^{-4}
FloydWarshall	adaptive	seq2seq	**0.00292**	0.00266	0.0025	0.00252	0.00268	0.00248	0.00262	0.0026	-2.0×10^{-4}
FloydWarshall	tropical	pointer	0.0063	0.00666	**0.00722**	0.00666	0.006	0.0057	0.00696	0.00608	5.0×10^{-4}
FloydWarshall	tropical	seq2seq	**0.00866**	0.00692	0.00656	0.00728	0.00768	0.00632	0.00678	0.00726	-0.0032
FloydWarshall	vanilla	pointer	0.00608	0.00618	0.00624	**0.00638**	0.00608	0.00632	0.00624	0.00628	1.0×10^{-4}
FloydWarshall	vanilla	seq2seq	0.00266	**0.00312**	0.00302	0.00282	0.00278	0.00298	0.00302	0.003	3.0×10^{-4}
FractionalKnapsack	adaptive	pointer	0.279	0.281	0.282	0.279	0.278	0.282	0.282	**0.285**	0.0019
FractionalKnapsack	adaptive	seq2seq	0.245	0.252	0.25	0.251	0.25	**0.254**	0.234	0.195	0.0077
FractionalKnapsack	tropical	pointer	0.309	0.312	0.312	0.313	0.314	**0.321**	0.309	0.308	0.0031
FractionalKnapsack	tropical	seq2seq	0.283	0.285	0.279	**0.297**	0.292	0.29	0.294	0.28	0.0024
FractionalKnapsack	vanilla	pointer	0.272	0.275	0.275	0.274	0.275	0.275	**0.279**	0.279	0.003
FractionalKnapsack	vanilla	seq2seq	**0.269**	0.267	0.266	0.265	0.264	0.25	0.237	0.221	-0.0016
Knapsack	adaptive	pointer	0.0722	0.0703	0.0685	0.0679	0.07	0.0708	0.0737	**0.0801**	-0.0019
Knapsack	adaptive	seq2seq	0.049	0.0514	0.0512	0.0468	0.0517	**0.0601**	0.0481	0.0229	0.0024
Knapsack	tropical	pointer	0.0896	0.0894	0.0909	**0.0927**	0.0856	0.0839	0.0839	0.0896	-1.6×10^{-4}
Knapsack	tropical	seq2seq	0.0499	0.069	0.0682	0.0655	0.0688	**0.0834**	0.0762	0.0532	0.019
Knapsack	vanilla	pointer	0.0633	0.0705	0.0677	**0.0729**	0.0693	0.0699	0.0655	0.0706	0.0072
Knapsack	vanilla	seq2seq	0.044	0.0483	0.0499	**0.0509**	0.049	0.0499	0.049	0.0258	0.0042
LIS	adaptive	pointer	**0.129**	0.127	0.125	0.123	0.114	0.122	0.119	0.113	-0.002
LIS	adaptive	seq2seq	0.0803	0.0769	0.0851	0.0745	0.0944	**0.0968**	0.0896	0.0637	-0.0033
LIS	tropical	pointer	**0.108**	0.0948	0.0894	0.1	0.107	0.0873	0.0868	0.0913	-0.014
LIS	tropical	seq2seq	0.049	**0.0616**	0.0557	0.0317	0.04	0.0532	0.0533	0.0339	0.013
LIS	vanilla	pointer	**0.0923**	0.0881	0.0863	0.0845	0.089	0.088	0.0873	0.0864	-0.0042
LIS	vanilla	seq2seq	0.0664	0.0918	**0.108**	0.0739	0.0891	0.0729	0.0626	0.0681	0.025
MaxSubsetSum	adaptive	pointer	**0.836**	0.818	0.818	0.818	0.818	0.818	0.819	0.817	0
MaxSubsetSum	adaptive	seq2seq	**1**	1	1	1	1	1	1	1	0
MaxSubsetSum	tropical	pointer	0.837	0.838	0.835	0.837	0.837	0.837	0.837	**0.838**	0
MaxSubsetSum	tropical	seq2seq	0.943	0.989	**0.989**	0.989	0.989	0.989	0.989	0.989	0.046
MaxSubsetSum	vanilla	pointer	**0.823**	0.819	0.818	0.819	0.82	0.819	0.817	0.816	0
MaxSubsetSum	vanilla	seq2seq	**1**	1	1	1	1	1	1	1	0
MinCoinChange	adaptive	pointer	0.121	0.121	0.121	0.121	0.121	0.121	0.122	**0.124**	-2.0×10^{-4}
MinCoinChange	adaptive	seq2seq	0.147	0.149	0.146	0.143	0.157	0.169	**0.209**	0.209	0.0018
MinCoinChange	tropical	pointer	0.152	0.145	0.15	0.151	0.151	0.149	0.145	**0.153**	-0.0064
MinCoinChange	tropical	seq2seq	0.153	0.153	0.147	0.148	0.142	0.171	**0.2**	0.2	-1.0×10^{-4}
MinCoinChange	vanilla	pointer	0.13	0.133	0.132	**0.133**	0.132	0.132	0.132	0.131	0.0033
MinCoinChange	vanilla	seq2seq	0.152	0.157	0.16	0.16	0.157	0.169	**0.201**	0.201	0.0053
Quickselect	adaptive	pointer	**0.167**	0.166	0.166	0.166	0.166	0.165	0.166	0.167	-9.6×10^{-4}
Quickselect	adaptive	seq2seq	0.178	0.174	0.173	0.173	0.174	0.174	0.176	**0.183**	-0.0048
Quickselect	tropical	pointer	0.231	0.228	0.236	0.223	0.234	**0.239**	0.234	0.219	-0.0028

Table 7: `ood_noise_exact` per campaign: means by `fishr_lamb`
Δ against 0.

campaign	model_type	head	0	0.01	0.1	1	10	100	1000	10000	Δ 0.01
Quickselect	tropical	seq2seq	0.155	0.154	0.153	0.154	0.151	0.159	0.174	**0.195**	-7.0×10^{-4}
Quickselect	vanilla	pointer	0.166	0.165	0.164	0.166	**0.166**	0.163	0.165	0.165	-6.6×10^{-4}
Quickselect	vanilla	seq2seq	0.181	0.179	0.18	0.178	0.178	0.178	0.178	**0.185**	-0.0026
SCC	adaptive	pointer	**0.0717**	0.0401	0.0366	0.041	0.0367	0.0355	0.0437	0.045	-0.032
SCC	adaptive	seq2seq	**0.0571**	0.0488	0.0442	0.0465	0.0489	0.0456	0.0468	0.0344	-0.0083
SCC	tropical	pointer	0.208	0.295	**0.418**	0.234	0.144	0.231	0.183	0.185	0.087
SCC	tropical	seq2seq	0.187	0.112	0.096	0.242	**0.323**	0.152	0.0494	0.0988	-0.075
SCC	vanilla	pointer	0.0442	0.063	0.0475	0.0466	0.0486	0.0637	0.0506	**0.0713**	0.019
SCC	vanilla	seq2seq	0.0444	0.0637	0.0624	0.0578	0.0536	0.0549	**0.0657**	0.0549	0.019
SubsetSumDecision	adaptive	pointer	**0.268**	0.23	0.239	0.228	0.233	0.238	0.237	0.251	-0.038
SubsetSumDecision	adaptive	seq2seq	0.496	0.504	0.499	0.505	0.481	0.483	0.515	**0.52**	0.0088
SubsetSumDecision	tropical	pointer	**0.211**	0.179	0.168	0.179	0.175	0.179	0.192	0.177	-0.032
SubsetSumDecision	tropical	seq2seq	0.242	0.256	0.366	0.329	0.257	0.233	0.321	**0.499**	0.014
SubsetSumDecision	vanilla	pointer	0.225	0.23	0.234	0.226	0.216	0.238	**0.243**	0.242	0.0042
SubsetSumDecision	vanilla	seq2seq	0.49	0.463	0.454	0.481	0.49	0.491	**0.507**	0.454	-0.027
ThreeSumDecision	adaptive	pointer	0.856	0.862	0.862	0.863	0.861	0.862	**0.863**	0.862	0.0058
ThreeSumDecision	adaptive	seq2seq	0.88	**0.88**	0.88	0.88	0.88	0.88	0.88	0.874	0
ThreeSumDecision	tropical	pointer	0.879	0.879	0.879	0.879	0.877	**0.879**	0.879	0.878	-5.0×10^{-4}
ThreeSumDecision	tropical	seq2seq	**0.88**	0.879	0.879	0.873	0.876	0.874	0.879	0.879	-2.2×10^{-4}
ThreeSumDecision	vanilla	pointer	0.848	0.853	0.853	0.852	0.852	0.852	0.853	**0.854**	0.0048
ThreeSumDecision	vanilla	seq2seq	0.88	0.88	**0.88**	0.88	0.88	0.88	0.879	0.876	0

A.6 `id_f1` (higher is better)

Table 8: `id_f1` per campaign: means by `fishr_la`
0.

campaign	model_type	head	0	0.01	0.1	1	10	100	1000
BalancedPartition	adaptive	pointer	0.791	0.79	0.791	0.79	0.791	0.791	**0.792**
BalancedPartition	adaptive	seq2seq	0.768	0.77	0.77	0.771	**0.771**	0.766	0.769
BalancedPartition	tropical	pointer	0.803	0.807	0.801	–	0.804	–	**0.816**
BalancedPartition	tropical	seq2seq	0.755	0.756	0.754	0.757	0.757	**0.758**	0.757
BalancedPartition	vanilla	pointer	0.778	0.778	0.777	0.776	0.778	0.777	0.778
BalancedPartition	vanilla	seq2seq	0.768	0.766	**0.769**	0.768	0.768	0.765	0.764
BinPacking	adaptive	pointer	**0.945**	0.944	0.942	0.943	0.945	0.94	0.93
BinPacking	adaptive	seq2seq	0.903	0.901	**0.909**	0.902	0.903	0.909	0.903
BinPacking	tropical	pointer	0.976	–	–	**0.984**	0.977	0.967	0.977
BinPacking	tropical	seq2seq	0.902	0.901	0.903	0.907	0.889	0.899	0.895
BinPacking	vanilla	pointer	0.939	0.941	0.94	0.94	**0.941**	0.935	0.928
BinPacking	vanilla	seq2seq	**0.914**	0.904	0.891	0.891	0.883	0.888	0.896
ConvexHull	adaptive	pointer	0.968	0.968	0.968	0.968	0.968	0.968	**0.968**
ConvexHull	adaptive	seq2seq	0.969	0.969	**0.969**	0.969	0.969	0.969	0.965
ConvexHull	tropical	pointer	0.984	0.985	0.985	0.985	**0.985**	0.984	0.985
ConvexHull	tropical	seq2seq	0.985	0.986	**0.986**	0.986	0.986	0.986	0.983
ConvexHull	vanilla	pointer	0.967	0.967	0.967	**0.967**	0.967	0.967	0.967
ConvexHull	vanilla	seq2seq	0.966	0.966	0.966	0.966	**0.966**	0.966	0.962
FloydWarshall	adaptive	pointer	0.954	0.954	0.954	0.954	0.954	0.954	0.954
FloydWarshall	adaptive	seq2seq	0.946	0.947	0.947	0.947	0.947	0.948	0.949
FloydWarshall	tropical	pointer	0.955	0.955	0.955	0.955	**0.956**	0.955	0.955
FloydWarshall	tropical	seq2seq	0.951	**0.952**	0.951	0.951	0.951	0.951	0.951
FloydWarshall	vanilla	pointer	0.954	0.955	0.955	0.955	0.955	0.955	0.955
FloydWarshall	vanilla	seq2seq	0.951	0.949	0.949	0.949	0.949	0.949	0.953
FractionalKnapsack	adaptive	pointer	0.949	0.949	0.95	0.949	0.949	**0.95**	0.949
FractionalKnapsack	adaptive	seq2seq	0.925	0.926	**0.926**	0.926	0.926	0.926	0.924
FractionalKnapsack	tropical	pointer	0.958	0.959	0.959	0.959	0.959	0.959	**0.959**
FractionalKnapsack	tropical	seq2seq	0.946	0.946	0.946	0.947	**0.947**	0.947	0.944

Table 8: id_f1 per campaign: means by fishr_la
0.

campaign	model_type	head	0	0.01	0.1	1	10	100	1000
FractionalKnapsack	vanilla	pointer	**0.944**	0.943	0.943	0.943	0.943	0.943	0.944
FractionalKnapsack	vanilla	seq2seq	0.924	0.924	0.924	**0.925**	0.924	0.923	0.922
Knapsack	adaptive	pointer	0.881	0.883	**0.883**	0.883	0.883	0.882	0.883
Knapsack	adaptive	seq2seq	0.873	0.873	**0.873**	0.873	0.873	0.873	0.871
Knapsack	tropical	pointer	0.88	0.882	0.882	0.882	0.881	0.882	0.882
Knapsack	tropical	seq2seq	0.881	**0.882**	0.882	0.881	0.881	0.881	0.879
Knapsack	vanilla	pointer	0.877	0.878	0.877	0.877	0.878	0.878	**0.878**
Knapsack	vanilla	seq2seq	0.872	0.873	0.873	0.873	0.873	**0.873**	0.871
LIS	adaptive	pointer	0.847	0.851	0.851	0.851	0.851	0.852	0.853
LIS	adaptive	seq2seq	0.783	0.783	0.783	0.783	0.783	**0.784**	0.782
LIS	tropical	pointer	**0.927**	0.922	0.925	0.922	0.921	0.919	0.919
LIS	tropical	seq2seq	0.844	0.855	**0.856**	0.853	0.851	0.849	0.831
LIS	vanilla	pointer	0.852	0.855	0.856	0.855	0.855	0.855	0.856
LIS	vanilla	seq2seq	0.793	**0.794**	0.793	0.793	0.793	0.79	0.787
MaxSubsetSum	adaptive	pointer	**0.986**	0.986	0.986	0.986	0.986	0.986	0.986
MaxSubsetSum	adaptive	seq2seq	**1**	1	1	1	1	1	1
MaxSubsetSum	tropical	pointer	0.986	**0.986**	0.986	0.986	0.986	0.986	0.986
MaxSubsetSum	tropical	seq2seq	**1**	1	1	1	1	1	1
MaxSubsetSum	vanilla	pointer	**0.986**	0.986	0.986	0.986	0.986	0.986	0.986
MaxSubsetSum	vanilla	seq2seq	**1**	1	1	1	1	1	1
MinCoinChange	adaptive	pointer	0.483	0.493	0.494	**0.496**	0.493	0.493	0.494
MinCoinChange	adaptive	seq2seq	0.397	0.397	0.401	**0.409**	0.38	0.275	0.194
MinCoinChange	tropical	pointer	0.57	0.573	**0.582**	0.582	0.579	0.58	0.568
MinCoinChange	tropical	seq2seq	0.446	**0.484**	0.482	0.447	0.44	0.359	0.242
MinCoinChange	vanilla	pointer	0.469	0.471	0.471	0.471	0.472	0.472	0.473
MinCoinChange	vanilla	seq2seq	0.421	0.421	0.42	**0.428**	0.422	0.291	0.262
Quickselect	adaptive	pointer	0.998	0.998	0.998	0.998	0.998	0.998	0.998
Quickselect	adaptive	seq2seq	0.939	0.939	0.939	**0.939**	0.939	0.938	0.934
Quickselect	tropical	pointer	0.943	0.946	0.947	0.945	0.945	0.945	**0.947**
Quickselect	tropical	seq2seq	0.752	0.748	0.752	0.749	**0.755**	0.742	0.723
Quickselect	vanilla	pointer	0.997	0.997	0.997	0.997	0.997	0.997	0.997
Quickselect	vanilla	seq2seq	0.941	0.941	0.941	**0.942**	0.942	0.941	0.933
SCC	adaptive	pointer	0.998	0.998	0.998	0.998	**0.998**	0.998	0.998
SCC	adaptive	seq2seq	0.998	0.998	0.998	0.998	0.998	0.998	**0.998**
SCC	tropical	pointer	0.998	0.998	0.997	0.998	0.998	0.997	**0.998**
SCC	tropical	seq2seq	0.999	0.999	0.999	0.999	0.998	0.998	**0.999**
SCC	vanilla	pointer	0.998	0.998	0.998	**0.998**	0.998	0.998	0.998
SCC	vanilla	seq2seq	0.998	**0.998**	0.998	0.998	0.998	0.998	0.998
SubsetSumDecision	adaptive	pointer	0.514	0.521	0.519	0.52	0.521	0.52	**0.522**
SubsetSumDecision	adaptive	seq2seq	0.382	**0.385**	0.382	0.381	0.382	0.372	0.263
SubsetSumDecision	tropical	pointer	0.556	0.569	0.561	**0.57**	0.564	0.567	0.566
SubsetSumDecision	tropical	seq2seq	**0.503**	0.499	0.502	0.495	0.5	0.493	0.432
SubsetSumDecision	vanilla	pointer	0.507	0.518	0.518	0.518	0.517	0.517	0.517
SubsetSumDecision	vanilla	seq2seq	0.384	0.383	0.381	0.383	**0.389**	0.383	0.276
ThreeSumDecision	adaptive	pointer	0.0183	0.0185	0.0165	**0.0221**	0.0203	0.017	0.0187
ThreeSumDecision	adaptive	seq2seq	2.20×10^{-4}	0	1.10×10^{-4}	0	3.30×10^{-4}	4.40×10^{-4}	0.00183
ThreeSumDecision	tropical	pointer	0.00346	0.00496	**0.00665**	0.00637	0.00549	0.00528	0.00593
ThreeSumDecision	tropical	seq2seq	0	**0.00487**	6.33×10^{-4}	0.00399	0.00377	0	3.17×10^{-4}
ThreeSumDecision	vanilla	pointer	0.0196	0.0239	0.0236	0.0241	0.0236	**0.0245**	0.0241
ThreeSumDecision	vanilla	seq2seq	2.20×10^{-4}	3.30×10^{-4}	4.40×10^{-4}	8.70×10^{-4}	3.30×10^{-4}	6.56×10^{-4}	0.00159

A.7 id_exact (higher is better)

Table 9: `id_exact` per campaign: means by `fishr_lambda`, and Δ agains
0.

campaign	`model_type`	`head`	0	0.01	0.1	1	10	100	1000	10000	Δ 0.01	Δ 0.1
BalancedPartition	adaptive	pointer	0.197	0.192	0.192	0.191	0.193	0.195	**0.197**	0.192	-0.005	-0.0045
BalancedPartition	adaptive	seq2seq	0.108	0.12	**0.12**	0.117	0.115	0.104	0.111	0.0939	0.012	0.012
BalancedPartition	tropical	pointer	0.254	0.267	0.247	–	0.257	–	**0.285**	–	0.012	-0.0072
BalancedPartition	tropical	seq2seq	0.0843	0.0875	0.0846	0.085	0.0861	**0.0882**	0.0827	0.0819	0.0032	3.4×10^{-4}
BalancedPartition	vanilla	pointer	**0.191**	0.185	0.186	0.183	0.186	0.187	0.187	0.189	-0.0051	-0.0047
BalancedPartition	vanilla	seq2seq	0.0969	0.0984	**0.101**	0.1	0.1	0.0956	0.0876	0.0799	0.0015	0.0044
BinPacking	adaptive	pointer	0.783	0.78	0.774	0.775	**0.785**	0.764	0.73	0.682	-0.003	-0.0087
BinPacking	adaptive	seq2seq	0.62	0.619	0.644	0.626	0.624	**0.644**	0.624	0.619	-0.0014	0.024
BinPacking	tropical	pointer	0.905	–	–	**0.934**	0.909	0.872	0.906	0.905	–	–
BinPacking	tropical	seq2seq	0.616	0.61	0.628	0.642	0.58	0.608	0.592	**0.647**	-0.0062	0.011
BinPacking	vanilla	pointer	0.758	0.766	0.762	0.761	**0.766**	0.745	0.715	0.664	0.0071	0.0033
BinPacking	vanilla	seq2seq	**0.663**	0.636	0.589	0.598	0.572	0.589	0.605	0.608	-0.027	-0.074
ConvexHull	adaptive	pointer	0.746	0.745	0.745	0.745	0.744	0.746	**0.747**	0.743	-8.8×10^{-4}	-5.0×10^{-4}
ConvexHull	adaptive	seq2seq	0.754	0.758	**0.758**	0.757	0.757	0.756	0.728	0.66	0.0037	0.0041
ConvexHull	tropical	pointer	0.864	0.869	0.869	0.869	0.87	0.868	**0.871**	0.868	0.0046	0.0051
ConvexHull	tropical	seq2seq	0.873	0.882	0.884	0.881	**0.884**	0.882	0.854	0.697	0.0088	0.011
ConvexHull	vanilla	pointer	0.736	0.736	0.736	**0.736**	0.736	0.736	0.736	0.736	-1.0×10^{-4}	2.0×10^{-4}
ConvexHull	vanilla	seq2seq	0.735	0.735	0.735	0.735	**0.736**	0.734	0.706	0.62	-2.0×10^{-4}	-2.6×10^{-4}
FloydWarshall	adaptive	pointer	0.622	0.621	0.622	0.622	0.623	0.622	0.622	**0.624**	-5.8×10^{-4}	4.0×10^{-4}
FloydWarshall	adaptive	seq2seq	0.588	0.591	0.591	0.591	0.591	0.595	0.599	**0.618**	0.0028	0.0029
FloydWarshall	tropical	pointer	**0.626**	0.623	0.621	0.624	0.625	0.622	0.624	0.62	-0.003	-0.0049
FloydWarshall	tropical	seq2seq	0.599	0.603	0.603	0.603	0.602	**0.604**	0.602	0.601	0.0046	0.0042
FloydWarshall	vanilla	pointer	0.622	0.624	0.624	0.624	0.624	0.624	0.623	**0.624**	0.0017	0.0017
FloydWarshall	vanilla	seq2seq	0.606	0.598	0.599	0.598	0.599	0.599	0.613	**0.622**	-0.0079	-0.0066
FractionalKnapsack	adaptive	pointer	0.632	0.637	0.637	0.636	0.636	**0.638**	0.635	0.626	0.0049	0.005
FractionalKnapsack	adaptive	seq2seq	0.5	0.509	**0.51**	0.509	0.508	0.509	0.498	0.454	0.0086	0.01
FractionalKnapsack	tropical	pointer	0.689	0.696	0.699	0.697	0.7	0.697	**0.701**	0.697	0.0075	0.01
FractionalKnapsack	tropical	seq2seq	0.623	0.623	0.623	0.625	**0.626**	0.625	0.609	0.536	-1.4×10^{-4}	-5.0×10^{-4}
FractionalKnapsack	vanilla	pointer	**0.602**	0.597	0.598	0.598	0.597	0.599	0.6	0.6	-0.0049	-0.0038
FractionalKnapsack	vanilla	seq2seq	0.484	0.493	0.492	**0.494**	0.492	0.487	0.48	0.437	0.0085	0.0075
Knapsack	adaptive	pointer	0.439	0.444	**0.445**	0.444	0.445	0.443	0.445	0.445	0.0052	0.0062
Knapsack	adaptive	seq2seq	0.394	**0.395**	0.395	0.393	0.394	0.393	0.388	0.361	4.4×10^{-4}	2.8×10^{-4}
Knapsack	tropical	pointer	0.438	0.446	0.446	0.446	0.445	0.446	0.445	**0.447**	0.0082	0.0081
Knapsack	tropical	seq2seq	0.438	**0.445**	0.444	0.441	0.441	0.439	0.432	0.393	0.0067	0.0053
Knapsack	vanilla	pointer	0.415	0.418	0.418	0.417	0.418	0.418	**0.42**	0.42	0.0029	0.0023
Knapsack	vanilla	seq2seq	0.39	0.394	0.394	0.394	0.394	**0.395**	0.388	0.357	0.0033	0.0032
LIS	adaptive	pointer	0.518	0.531	0.531	0.531	0.53	0.531	0.534	**0.536**	0.013	0.013
LIS	adaptive	seq2seq	0.313	0.314	0.314	0.313	0.314	**0.317**	0.308	0.279	0.0012	0.0014
LIS	tropical	pointer	**0.769**	0.754	0.768	0.753	0.755	0.746	0.75	0.745	-0.015	-6.6×10^{-4}
LIS	tropical	seq2seq	0.509	0.532	0.531	**0.533**	0.531	0.52	0.484	0.417	0.023	0.022
LIS	vanilla	pointer	0.522	0.532	0.533	0.533	0.533	0.533	0.533	**0.541**	0.01	0.011
LIS	vanilla	seq2seq	0.347	0.356	0.356	**0.357**	0.353	0.336	0.316	0.282	0.0093	0.0093
MaxSubsetSum	adaptive	pointer	0.984	0.984	0.984	**0.984**	0.984	0.984	0.984	0.984	0	0
MaxSubsetSum	adaptive	seq2seq	**1**	1	1	1	1	1	1	1	0	0
MaxSubsetSum	tropical	pointer	0.984	**0.984**	0.984	0.984	0.984	0.984	0.984	0.984	0	0
MaxSubsetSum	tropical	seq2seq	**1**	1	1	1	1	1	1	1	0	0
MaxSubsetSum	vanilla	pointer	0.984	0.984	0.984	**0.984**	0.984	0.984	0.984	0.984	0	0
MaxSubsetSum	vanilla	seq2seq	**1**	1	1	1	1	1	1	1	0	0
MinCoinChange	adaptive	pointer	0.31	0.309	0.308	0.309	0.31	0.31	0.312	**0.315**	-0.0011	-0.0015
MinCoinChange	adaptive	seq2seq	0.268	**0.269**	0.268	0.261	0.244	0.232	0.213	0.213	0.0016	5.6×10^{-4}
MinCoinChange	tropical	pointer	0.42	0.419	0.431	**0.437**	0.428	0.427	0.416	0.427	-0.0013	0.01
MinCoinChange	tropical	seq2seq	0.298	**0.333**	0.311	0.3	0.294	0.247	0.204	0.204	0.035	0.013
MinCoinChange	vanilla	pointer	0.302	0.302	0.302	0.303	0.302	0.303	0.302	**0.304**	4.8×10^{-4}	0
MinCoinChange	vanilla	seq2seq	0.268	0.271	0.275	**0.275**	0.26	0.245	0.2	0.2	0.003	0.0062
Quickselect	adaptive	pointer	0.995	0.995	0.995	0.995	0.995	0.995	**0.995**	0.995	1.0×10^{-4}	-3.0×10^{-4}
Quickselect	adaptive	seq2seq	0.866	0.866	0.866	**0.866**	0.866	0.865	0.86	0.836	-4.0×10^{-4}	-5.0×10^{-4}
Quickselect	tropical	pointer	0.865	0.87	0.872	0.866	0.868	0.867	**0.872**	0.871	0.0035	0.0062

Table 9: `id_exact` per campaign: means by `fishr_lambda`, and Δ agains
0.

campaign	`model_type`	`head`	0	0.01	0.1	1	10	100	1000	10000	Δ 0.01	Δ 0.1
Quickselect	tropical	seq2seq	**0.62**	0.605	0.603	0.617	0.613	0.595	0.569	0.512	-0.0098	-0.018
Quickselect	vanilla	pointer	0.993	0.994	0.994	0.994	0.994	0.994	0.994	**0.994**	8.0×10^{-4}	0.0012
Quickselect	vanilla	seq2seq	0.87	0.871	0.871	**0.872**	0.872	0.87	0.858	0.83	0.0019	0.0018
SCC	adaptive	pointer	**0.985**	0.985	0.985	0.985	0.985	0.985	0.985	0.985	-9.0×10^{-4}	-6.0×10^{-4}
SCC	adaptive	seq2seq	0.984	0.988	0.988	0.988	0.988	0.989	**0.989**	0.988	0.0028	0.0036
SCC	tropical	pointer	0.986	0.988	0.985	0.987	0.987	0.985	0.987	**0.988**	-5.0×10^{-4}	0.0011
SCC	tropical	seq2seq	**0.995**	0.993	0.992	0.994	0.992	0.99	0.994	0.994	-1.0×10^{-4}	-3.0×10^{-4}
SCC	vanilla	pointer	0.983	0.985	**0.985**	0.985	0.985	0.985	0.985	0.985	4.0×10^{-4}	6.0×10^{-4}
SCC	vanilla	seq2seq	0.987	0.99	0.99	**0.99**	0.99	0.99	0.99	0.989	0.0037	0.0035
SubsetSumDecision	adaptive	pointer	0.572	0.572	0.573	0.572	0.572	0.573	0.573	**0.575**	-3.4×10^{-4}	4.0×10^{-4}
SubsetSumDecision	adaptive	seq2seq	0.626	0.626	0.626	**0.628**	0.628	0.623	0.585	0.527	5.8×10^{-4}	2.2×10^{-4}
SubsetSumDecision	tropical	pointer	0.627	**0.63**	0.621	0.626	0.621	0.625	0.624	0.619	0.003	-0.0063
SubsetSumDecision	tropical	seq2seq	0.637	0.635	0.636	0.636	**0.643**	0.642	0.617	0.549	-0.0022	-5.4×10^{-4}
SubsetSumDecision	vanilla	pointer	0.575	0.573	0.573	0.573	0.573	0.573	0.573	**0.576**	-0.0022	-0.0018
SubsetSumDecision	vanilla	seq2seq	0.625	0.627	**0.628**	0.628	0.627	0.625	0.588	0.525	0.0016	0.0029
ThreeSumDecision	adaptive	pointer	0.872	0.872	**0.872**	0.871	0.872	0.872	0.872	0.871	-3.2×10^{-4}	8.0×10^{-5}
ThreeSumDecision	adaptive	seq2seq	0.876	**0.876**	0.876	0.876	0.876	0.876	0.876	0.874	4.0×10^{-5}	2.0×10^{-5}
ThreeSumDecision	tropical	pointer	**0.875**	0.875	0.875	0.875	0.875	0.875	0.875	0.875	-4.2×10^{-4}	-6.0×10^{-4}
ThreeSumDecision	tropical	seq2seq	**0.876**	0.876	0.876	0.876	0.876	0.876	0.876	0.874	-2.8×10^{-4}	-2.0×10^{-5}
ThreeSumDecision	vanilla	pointer	**0.872**	0.872	0.871	0.872	0.872	0.872	0.872	0.872	-4.6×10^{-4}	-6.2×10^{-4}
ThreeSumDecision	vanilla	seq2seq	0.876	0.876	0.876	0.876	**0.876**	0.876	0.876	0.874	-4.0×10^{-5}	-6.0×10^{-5}